SG No. 112 D_{2d}^{2} $P\bar{4}2c$ [tetragonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `222.`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[3,4,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[3,4,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `222.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,7,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,6]`

continued ...

Table 7

No.	position	mapping
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[3,7]`
4	$\left[0,\ x,\ \frac{3}{4}\right]$	`[4,8]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[y,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,8]`
4	$\left[-y,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,7]`

Table 9: Wyckoff site: `4i`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[3,7]`
4	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[4,8]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[2,6]`
3	$\left[y,\ 0,\ \frac{3}{4}\right]$	`[3,8]`
4	$\left[-y,\ 0,\ \frac{3}{4}\right]$	`[4,7]`

Table 11: Wyckoff site: `4k`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$[0,\ 0,\ -z]$	`[3,4]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,8]`

Table 12: Wyckoff site: `4l`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,4]`

continued ...

Table 12

No.	position	mapping
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 13: Wyckoff site: `4m`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[7,8]`

Table 14: Wyckoff site: `8n`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[y,\ -x,\ -z]$	`[3]`
4	$[-y,\ x,\ -z]$	`[4]`
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[6]`
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[8]`